

Design to Improve Social Presence among University Students in Asynchronous Online Learning Environment

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ABSTRACT

Since the COVID-19 pandemic, online lectures are becoming more common in higher education. Specifically, asynchronous online classes have become increasingly popular because of their flexibility. Asynchronous online courses, however, may negatively impact students' academic performance and social development due to the diminished sense of social presence. To explore ways to enhance social presence among students in asynchronous online classes, this paper used a co-design methodology that involved 12 undergraduate students as primary stakeholders. As a result, we developed a design framework for designing in-class interaction to promote social presence in asynchronous online lectures. This framework consists of four high-level elements and sub-categories: interaction topic (direct or peripheral topics related to learning), interaction size (small or entire group), interaction mode (anonymity, synchronicity, instructor involvement), and interaction motivator (lightweightness and entertainment). Our design framework may serve as a guide to future technology for improving asynchronous online classes.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative and social computing**.

KEYWORDS

Social presence, university education, higher education, online learning, asynchronous learning, asynchronous online class, co-design

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1 RESEARCH BACKGROUND

As a result of the COVID-19 pandemic, many universities have converted their face-to-face classes into online classes [12]. Online courses at universities usually take the form of synchronous classes using video conferencing programs like Zoom and asynchronous classes using recorded videos. Asynchronous online classes, in particular, provide greater flexibility for students as they can access classes without space or time limitations, rather than having to be logged in at a specific time for synchronous online classes. This can be especially beneficial for students who have other responsibilities, such as work or caregiving, that make it difficult for them to attend classes at set times. However, this flexibility can also result in a reduction in social presence and opportunities to interact with other students and instructors.

Social presence, which is “the subjective feeling of being with others” [4], can be easily felt in traditional face-to-face class settings where the presence of others is present and active interaction can take place [17]. In the context of higher education, social presence can be understood as the extent to which students feel connected to their peers and instructors, feel part of a university community, and believe that their contributions are valued and regarded [5, 14]. In this sense, social presence has been shown to increase learners' cohesion, satisfaction, and interaction in online learning, and have a positive effect on academic performance [2, 16]. However, the fuller sense of social presence can be difficult to simulate online, especially for asynchronous online classes. With students feeling isolated and lacking the opportunity to feel the presence of classmates and instructors, research revealed that students tend to be less motivated, and they are more likely to skip classes [6].

Even after the COVID-19 pandemic, digital transformations in education are likely to continue, as asynchronous online lectures can provide greater flexibility, accessibility, and convenience for

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students and can also allow for greater scalability for universities [19]. However, the purpose of higher education is not only limited to academic goals but also involves social development goals, where they provide students with opportunities to interact with their peers, form friendships, and build a sense of community [20].

In order for an asynchronous online class to meet both academic and social development goals, it is essential to promote social presence among students. While previous literature highlighted the importance of social presence for online learning environments [23], there are relatively few studies on how to improve social presence, particularly for asynchronous online classes. In most previous studies, the main focus was on synchronous online classes, such as simulating face-to-face classes through immersive virtual environments (e.g., [8]).

In this paper, we investigated ways to increase social presence of asynchronous online lectures in higher education settings using a co-design methodology. A co-design process involves stakeholders participating in the design process, helping to identify and resolve problems collaboratively [22]. In order to integrate stakeholder perspectives and experiences into the design of future learning platforms, We recruited 12 undergraduate students who had experienced asynchronous online lectures and felt reduced social presence in those classes. Three sessions of collaborative ideation and sketches generated novel design ideas.

By observing co-design sessions and analyzing participants' design ideas, we developed a design framework for designing in-class interaction to promote social presence in asynchronous online lectures. This framework consists of four high-level elements and sub-categories: interaction topic (direct or peripheral topics related to learning), interaction size (small or entire group), interaction mode (anonymity, synchronicity, instructor involvement), and interaction motivator (lightweightness and entertainment). The framework may serve as a guide to future technology for improving asynchronous online classes.

2 RESEARCH METHOD

2.1 Participants

As stakeholders of the research problem, undergraduate students were recruited who attended at least one asynchronous online lecture at their university during the COVID-19 pandemic lockdown. Particularly, we focused on recruiting those who felt reduced social presence in asynchronous online classes, since we saw that they could meaningfully relate to our co-design process.

Following approval from the Institutional Review Board (IRB) of the university where the study was hosted, recruitment posts were posted on online social communities for university students. We asked those interested in participating to fill out an online questionnaire. The questionnaire asked about demographic information, experience taking asynchronous online lectures, and the sense of social presence in those classes. The sense of social presence was measured on a 5-point Likert scale using the social presence scale for students' involvement in online learning [9]. In the previous study [18], if the average response of the items was 3.45 or less, it was considered to have reduced sense of social presence. Following these criteria, a total of 12 participants were selected (7 female,

average age: 22.7 years old). Participants received 30,000 Korean Won (equivalent to \$24 USD) for their participation.

2.2 Procedures

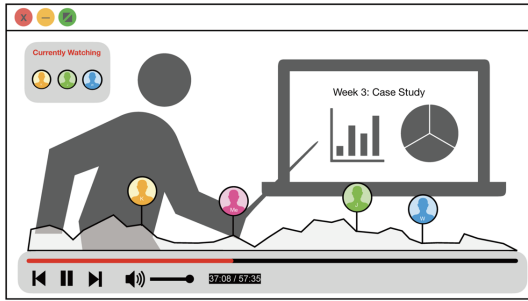
A total of three co-design sessions were conducted in person. Each session was attended by four participants, with three researchers facilitating the session and taking notes. Each session lasted about two hours and was audio-recorded with the consent of the participants. The procedure of co-design sessions was as follows.

- **Warm up (10 minutes):** After receiving consent for participation and audio recording, participants and researchers introduced themselves. A detailed explanation of the topic, goal, and procedures of the session was provided by the researchers.
- **Focus Group Discussion (20 minutes):** During the focus group discussion, researchers encouraged participants to share their experiences and problems attending asynchronous online lectures with a focus on social presence. After then, researchers guided participants in choosing the common issues of asynchronous online classes for the following ideation and sketching activities.
- **Individual Brainstorming (10 minutes):** In order to generate solution ideas, we introduced Crazy 8's [21], an ideation tool in which eight sketches are drawn in eight minutes. By setting a time limit, the rapid generation of multiple ideas could be facilitated [10]. As a result, all participants generated multiple solutions ideas to the common problems they agreed on.
- **Group sketch (20 minutes):** The two subgroups (2 members each) were formed within each session and were asked to collaborate on a group design sketch collaboratively using individual idea sketches from individual brainstorming.
- **Presentation and Feedback (20 minutes):** Participants presented their group sketches and exchanged feedback. As a tool for the constructive feedback process, the Stickies method [25] was used to evaluate and improve group idea sketches based on the good points (Likes), the bad points (Dislikes), and the points to improve (Design Ideas).
- **Revision (10 minutes):** Using feedback for the group sketches, each group sketch was iteratively revised.
- **Wrap up (20 minutes):** After presenting the revised group sketches, participants discussed ways to enhance the social presence of the asynchronous online class. Following this, we compensated participants and concluded the session.

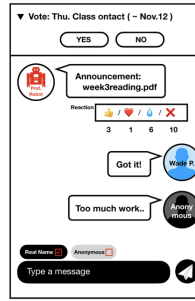
2.3 Data collection and Analysis

We transcribed the audio recordings of all sessions. Following that, we incorporated the individual sketches and group sketches into the transcripts for data analysis. Three researchers read the transcripts and open-coded them individually, which resulted in 725 codes and memos. In the next step, researchers converted all of them to digital sticky notes on Miro [13] for affinity diagramming [1]. As we iteratively organized and grouped relevant codes together using affinity diagramming, we identified 29 themes and consolidated them into four high-level themes (i.e., interaction topic, size, mode, and motivator). We then identified the sub-themes and categories

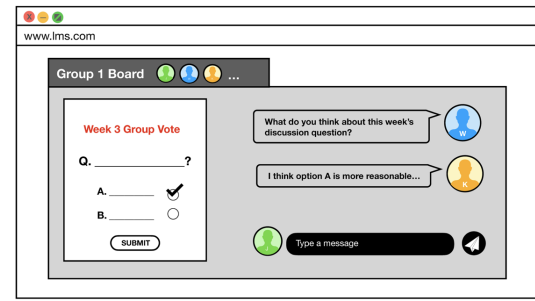
1. Study Run



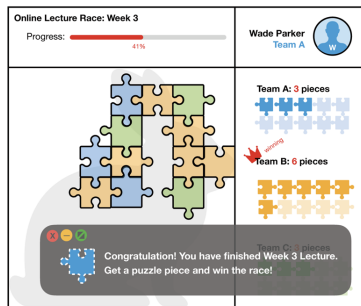
2. Professor-Free Chat



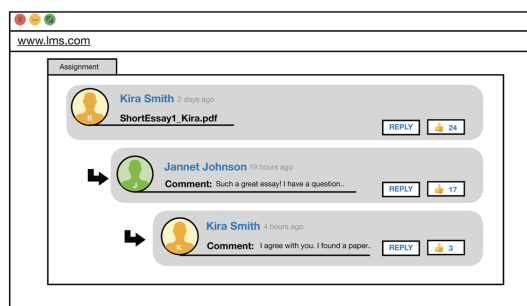
3. Study Group Board



4. Class Puzzle



5. Colo-gora



6. Lecture Talk

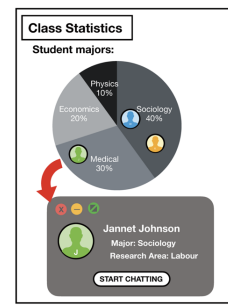


Figure 1: Design sketches from the co-design sessions (1. Study Run, 2. Professor-Free Chat, 3. Study Group Board, 4. Class Puzzle, 5. Colo-gora, 6. Lecture Talk)

and characterized the relations between them. After all, this analytic process allowed us to develop design frameworks and suggestions to improve social presence in asynchronous online classes.

3 FINDINGS

In the focus group discussion, all participants expressed a reduced sense of social presence in asynchronous online classes. P8 said that *"I had no expectation of social presence in asynchronous lectures"* and P5 shared that *"I didn't feel like I belonged at university."* Even so, some participants felt a sense of social presence at times, such as when *"my professor mentioned other students' questions in the lecture video"* (P8) and when *"another student answered my question via the online class bulletin board"* (P11).

Across all sessions, participants agreed that promoting in-class interaction among students was essential to increasing social presence in asynchronous online courses. Therefore, they envisioned various ways to enhance student interaction in their design ideas in the Co-design session. Overall, a total of 6 group sketches were created. Below, we described the main features of their designs. Group sketches on papers were digitized by researchers and included in Figure 1.

First, **'Study Run'** (created in the first session by P1 and P4) is the design idea that displays avatars of learners on the process bar of the lecture video player in real-time so that users can see who is currently watching the lecture video and who has finished it. Additionally, the lecture video statistics are displayed as a line graph

so users can view which sections of the videos their classmates have viewed the most.

'Professor-Free Chat' (created in the first session by P2 and P3) is a group chat interface in which students share their thoughts and reactions to class-related notices without the presence of a class instructor.

'Study Group Board' (created in the second session by P5 and P8) is a design whereby students are randomly divided into groups for discussion and voting on class content. The instructor collects the voting results and shares them in the class video afterward.

'Class Puzzle' (created in the second session by P6 and P7) is a design in which each student receives a puzzle piece when they finish a lecture video. During this activity, students are automatically divided into small groups, and the group with the most puzzle pieces at the end wins.

'Colo-gora' (created in the third session by P11 and P12) is a design in which students work on assignments and post them on a bulletin board, then give anonymous feedback on each other's work.

Lastly, **'Lecture Talk'** (created in the third session by P9 and P10) is a design that visualizes class statistics (e.g., average grades for assignments, progress on class materials, number of students by major) and allows students to discuss class statistics without instructors present.

Through the analysis of the design features of each sketch as well as transcripts of the session recordings, the design framework for

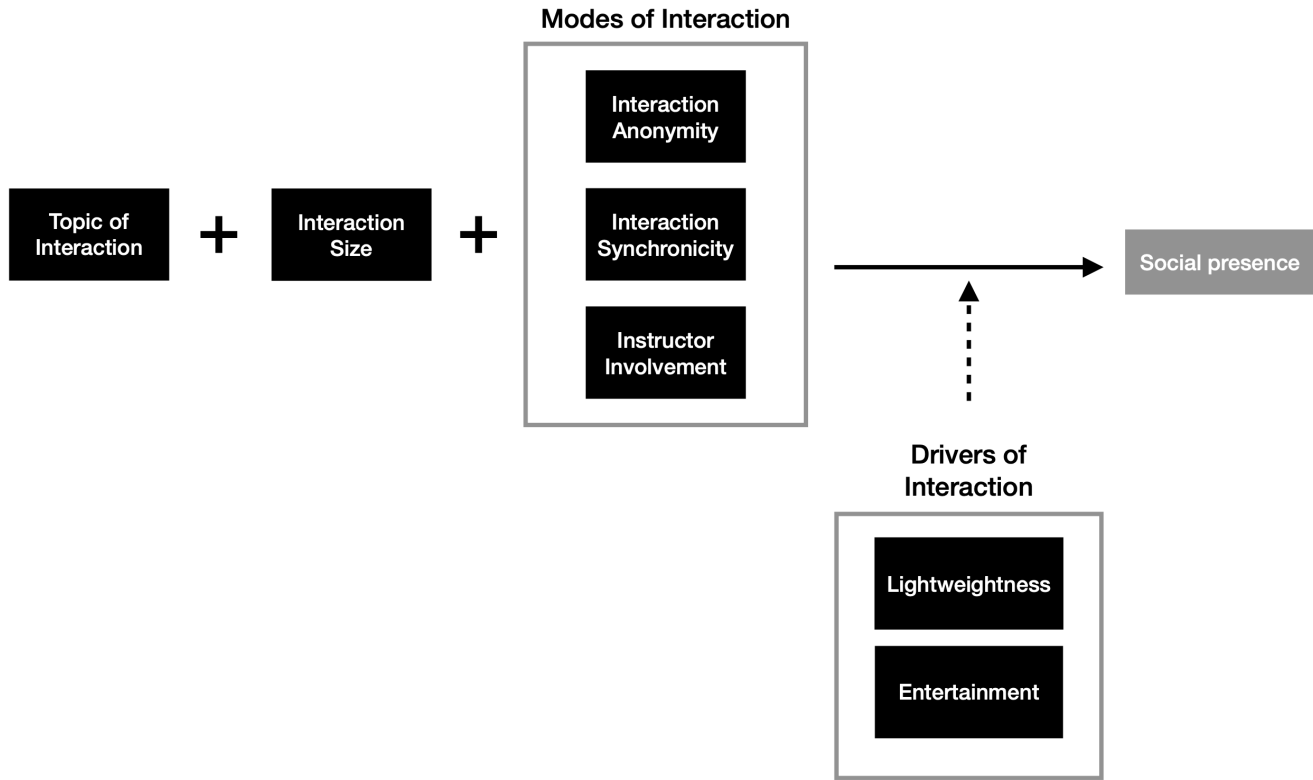


Figure 2: Design framework to design in-class interaction to promote social presence in asynchronous online classes

promoting in-class interaction and social presence among students in asynchronous online lectures was derived (see Figure 2).

As a **topic of interaction**, participants saw both direct topics related to their learning processes such as class materials and announcements, and peripheral topics such as class-wise attendance statistics helpful for enhancing social presence. Their preference for **interaction size** varied; some ideas expressed a desire to interact more with small groups and others to interact with the entire class. Also, different sketches reflected diverse functions of the **mode of interaction** in regard to **anonymity** (full vs. partial anonymity), **synchronicity** (synchronous vs. pseudo-synchronous interaction), and **instructor involvement** (instructor mediation vs. instructor-free interaction). Lastly, lightweightness and entertainment were emphasized as **drivers of interaction** to promote more student involvement.

3.1 Topic of Interaction

Some participants preferred topics **directly related to learning** such as class materials and class announcements. As a common ground for the social presence among classmates, they valued the opportunities to work together towards better learning outcomes as reflected in the design sketch of ‘Colo-gora’ (bulletin board for exchanging feedback on each other’s assignment). Also, they valued sharing thoughts and reactions regarding class announcements, as reflected in the idea of ‘Professor-Free Chat.’

Other ideas showed **peripheral topics**, which were not directly related to their learning processes such as class-wise statistics (e.g., attendance rates or grade distributions) could be effective topics for interaction (e.g., ‘Class Puzzle’, ‘Lecture Talk’). Despite not necessarily involving direct bi-directional components nor deep discussion of class materials, participants commented that having access to those statistics made them feel as if they were “*not taking the class alone*” (P1).

3.2 Interaction Size

Participants expressed varying preferences concerning the size of the group for in-class interaction to enhance their sense of social presence. Some ideas, such as ‘Study Group Board’ and ‘Class Puzzle’, suggested the preference for **small groups**. Developing a strong interpersonal relationship with smaller groups of classmates was considered an important factor in forming a social presence.

However, there was also a demand to establish a social presence for the **entire class**. The most common way of sharing learners’ information was through class statistics, such as lecture attendance rates and grade distributions (e.g., ‘Study Run’). P2 commented that even without direct interaction, there would be a sense of belonging if information about the whole was provided, such as the number of people watching recorded lecture videos. Also, some envisioned a group chat for the entire class, such as in ‘Professor-free Chat,’ would help them feel included as a part of the class.

3.3 Modes of Interaction

Participants discussed different modes of interaction in regard to **anonymity** (full vs. partial anonymity), **synchronicity** (synchronous vs. pseudo-synchronous interaction), and **instructor involvement** (instructor mediation vs. instructor-free interaction).

3.3.1 Interaction Anonymity. Anonymity was a common preference among participants in regard to increasing social presence. Because they felt burdened about exposing themselves online without knowing much about their classmates, they preferred anonymity: “I would avoid using my real name unless it was required by the instructor” (P8). In some design sketches, anonymity took the form of **full anonymity** (e.g., ‘Colo-gora’), where individuals could not be identified, and **partial anonymity**, where individuals or teams could be identified (e.g., ‘Study Run’, ‘Class Puzzle’).

In the scenario of full anonymity, despite not being able to identify each classmate, some saw they could still maintain a sense of social presence with their entire class. As a way to augment a sense of social presence through complete anonymity, they suggested displaying class data and statistics such as assignment submission rates, or allowing anonymous comments.

As with partial anonymity, it was still impossible to identify each individual classmate, but they could at least distinguish each from the other. For example, in the ‘Study Run’ sketch, each student was distinguished by different avatars. In ‘Class Puzzle’, puzzle colors were set differently by groups to distinguish between teams in the class. P1 saw that this partial anonymity could simulate similar senses of social presence in in-person classrooms: “It is almost as if we could roughly recognize classmates by their appearances [in face-to-face classroom settings], but do not know deeply about them.”

3.3.2 Interaction Synchronicity. Overall, participants saw increasing interaction synchronicity would enhance learners’ social presence. Even if they preferred the lecture itself to take place asynchronously, some participants hoped to have access to the most up-to-date class information and statistics **synchronously**: “I would feel as if I was listening together if I saw a real-time assignment submission rate or concurrent users increase” (P1). Further, **pseudo-synchronicity** has been proposed in order to give the illusion of real-time interaction. For example, several participants proposed allowing fellow learners to see comments posted at specific times above a recorded lecture video: “I would feel as if they were watching the video with me at the same time” (P3).

3.3.3 Instructor Involvement. There was also a discussion about the role of the instructor in creating a sense of social presence among students. There were opposing views on instructors’ role in promoting social presence. Some participants felt that **instructor involvement** was essential to promote a sense of social presence by facilitating discussion and interactive class activities (e.g., voting). Some thought that **instructor-free interaction** would help facilitate more active and natural interaction among students as reflected in the ‘Professor-Free Chat’. Further, P7 was concerned that “instructor intervention would decrease fun.”

3.4 Drivers of Interaction

In order to promote more active interaction among students in asynchronous lectures, two drivers of interaction are discussed: lightweightness and entertainment features.

3.4.1 Lightweightness. The lightweight nature of interaction was discussed as a key element to reduce students’ burdens and not to disrupt their learning process. Participants stressed the importance of integrating new interaction features into existing learning management interfaces without having to download or sign up for separate systems. Also, some respondents wanted social interaction materials automatically generated with minimal effort. For example, in the ‘Class Puzzle’ idea, individual puzzles were automatically generated and reflected after a video clip was completed by a student.

3.4.2 Entertainment. Participants also emphasized that entertainment features, such as competition, would drive them to “*actually take classes*” (P7). P4 mentioned that while such entertainment features had not been essential in the educational context, they would become more important in asynchronous online learning settings as it would help “*eliminate hesitation to participate in class interaction*”.

4 DISCUSSION

Through our findings, we developed a framework for promoting social presence in asynchronous online lectures (Figure 2). This framework contains high-level design elements to design in-class interaction (i.e., interaction topic, size, mode and driver) to promote social presence in asynchronous online lectures.

As a **topic of interaction**, participants saw both direct topics related to their learning processes such as class materials and activities, and peripheral topics such as class-wise attendance statistics helpful for enhancing social presence. While interaction around direct topics to learning was viewed as a way to improve their learning outcomes, peripheral topics were considered as a means for fostering empathy and a sense of community among students by reminding them of shared experiences. Previous research suggests that online learners are more focused on building knowledge than building relationships among them [11]. However, social aspects of higher education could be just as important as academic achievement [5, 14]. To enhance the feelings of community within asynchronous online classes, providing more peripheral information, such as class-specific statistics, may be helpful.

Also, we found that participants’ preference for **interaction size** varied. Some saw that small group interaction would allow them to have a more close and direct interaction with others, thus increasing their sense of social presence. However, some saw that indirect but overall sense of all the others in the class would be more feasible solutions for promoting social presence.

As for **modes of interaction**, participants generally preferred some level of **anonymity** while expressing a desire for more interaction with others. Previous research has shown that students often prefer to use pseudonyms or remain anonymous when participating in online discussions or other interactive learning activities with their peers [3, 15]. One reason for this is that anonymity can make students feel more comfortable sharing their ideas and opinions without fear of judgment or social pressure from their peers [7].

Aligned with findings from previous literature, our result suggests that many participants did not feel comfortable or natural participating in class interaction in asynchronous learning environments. For this reason, allowing some degree of anonymity would be a critical design consideration for promoting social interaction among students in asynchronous online classes.

Also, this general reluctance to class interaction in asynchronous classes may explain why participants emphasized **lightweightness** and **entertainment** as a **driver** to reduce hesitancy and burden for in-class interaction. Our findings suggested that lightweight interactions, such as ones that do not require extra cognitive processing, time, or tasks, could foster social interaction among students more effectively while not disrupting their learning processes much. Also, entertainment features such as lightweight competitions and gamification could be useful approaches in decreasing hesitation and reluctance in class interactions.

In addition, participants' ideas expressed a desire to have some level of **synchronicity** even in asynchronous lectures. Even though they appreciated the flexibility of asynchronous lectures, they saw design elements that could mediate the feeling of simultaneous interaction as helpful in helping them sense the presence of others. Diverse ways to mediate pseudo synchronicity [6, 8] have been experimented with massive open online course (MOOC) settings (e.g., [26]). For instance, by applying Danmaku, a form of video that shows overlay timeline-dependent comments to MOOC videos, learners can experience synchronicity and feel more connected to other learners [27]. Those approaches could be also applicable to asynchronous online courses in universities.

Further, we found that **instructor involvement** was a double-edged sword: one side argues that instructors are most responsible for facilitating student interaction, on the other, the presence of instructors can negatively impact student interactions. Similarly, previous research has shown that instructors' presence played an important role in helping learners initiate interaction; however, direct guidance negatively affected interactions between students [24]. Overall, this suggests that instructor involvement would be effective for the initial phase of interaction, but perhaps less so for the subsequent phase of fostering relationships and community among students.

As a whole, we characterized two types of social presence for asynchronous classes: a **micro** but strong sense of social presence among a small group of students compared to a **macro** but weak sense of social presence for the entire class.

Using our design framework, we anticipate that such micro presence can be achieved better when interaction size is small, partial anonymity is present, and instructor presence is minimal. These design features to promote a micro sense of social presence could help students develop meaningful relationships and a sense of connection among a small group of peers. Therefore, small class settings focusing on collaborative activities and discussion will be facilitated more effectively with this approach. In contrast, we expect that a macro sense of social presence can be achieved better when some topics of interaction are peripheral, the size of interaction is for the entire class, and full anonymity is allowed. This approach would be effective in large lecture-style classes with little need for direct interaction among students, fostering a sense of belonging among a large number of students.

5 CONCLUSION

Using a co-design methodology, this paper explored ways to increase learners' social presence in asynchronous online classes in universities. Our data analysis led us to propose a design framework that includes four categories: interaction topic, interaction size, interaction mode, and interaction driver. This framework hopes to provide a guide to promote in-class interaction and thus increase social presence in asynchronous online courses.

Although this paper offers valuable insights, it has some limitations since the sample of participants was small and included only participants who experienced a lack of social presence. As part of future studies, we intend to broaden our participant pool by involving a diverse range of stakeholders, including students, instructors, and university personnel. Further, we intend to refine and validate our design framework through an experimental study by implementing some of the participants' ideas into prototypes and implementing them in asynchronous courses.

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